NIOS Class 10 (212) Science & Technology — 7-Day Fast-Track Study Plan

NIOS Secondary (Class 10) Science & Technology (212) — Exam Prep Study Kit

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Designed for Working Students | Target: 55+ to 70+ Marks out of 85

> **Daily Time Commitment**: 2 hours on weekdays (45 mins morning + 1 hr 15 mins evening), 3–4 hours on weekend.

The Working Student's Daily Rhythm

- **Morning Sprint (45 mins)**: Read the Formula & Concept cheat sheet for that day's module. Memorize 3 key definitions or diagrams.
- **Commute / Breaks (15 mins)**: Review chemical formulas or SI units on your phone.
- **Evening Session (75 mins)**: Write down 2 diagrams, solve 1 physics numerical, and review 10 objective MCQs.

7-Day Fast-Track Roadmap

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graph TD
    D1["Day 1: High-Yield Physics Optics & Eye (15 Marks)<br>Mirror/Lens Formula + Ray Diagra
    D2 --> D3["Day 3 (Weekend): Biology Life Processes & Heredity (18 Marks)<br>Heart + Nephr
    D3 --> D4["Day 4 (Weekend): Chemistry Salts & Carbon (16 Marks)<br>4 Key Salts + Metals/N
    D4 --> D5["Day 5: Mechanics, Gravitation & Periodic Table (12 Marks)<br>F=ma + Gravitatio
    D5 --> D6["Day 6: Ecology & Top 50 Objectives Blitz (42 Marks)<br>10% Law + Food Chain +
    D6 --> D7["Day 7: Final Rehearsal & Mock Paper<br>Practice 1 Mock Test + Quick Glance at
```

Day 1: High-Yield Physics — Light & Human Eye (15 Marks)

- **Target Chapters**: L.15 (Light Energy)
- **Core Topics**:
- Reflection & Refraction: Snell's Law ($\frac{\sin i}{\sin r} = \mu$).
- Mirror Formula ($\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$) and Lens Formula ($\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$).
- Power of Lens: $P = \frac{1}{f \text{ (in metres)}}$ (Unit: Dioptre, D).
- Eye Defects: Myopia (corrected by concave lens) vs Hypermetropia (corrected by convex lens) with ray diagrams.
- **Practice**: Draw the two eye correction diagrams on paper!

Day 2: High-Yield Physics — Electricity & Magnetism (15 Marks)

- **Target Chapters**: L.16 (Electrical Energy) & L.17 (Magnetic Effect)
- **Core Topics**:
- Ohm's Law: $V = IR$. Circuit diagram for verification.
- Equivalent resistance: Series ($R_s = R_1 + R_2 + R_3$) vs Parallel ($\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$).
- Electric Power ($P = VI = I^2R = \frac{V^2}{R}$) and Commercial Energy billing ($1\ \mathrm{kWh} = 1\ \text{unit} = 3.6 \times 10^6\ \mathrm{J}$).
- Fleming's Left-Hand Rule (Electric Motor) & Domestic Circuit Safety (Earth wire, Fuse).
- **Practice**: Solve 2 numerical problems on equivalent resistance and energy cost.

Day 3 (Weekend): Biology Core — Life Processes & Heredity (18 Marks)

- **Target Chapters**: L.22 (Life Processes I), L.23 (Control), L.24 (Reproduction), L.25 (Heredity)
- **Core Topics**:
- Human Heart & Double Circulation (Why is oxygenated separated from deoxygenated blood?).
- Nephron: Ultrafiltration and urine formation.
- Aerobic vs Anaerobic respiration pathways.
- Mendel's Monohybrid Cross: $3:1$ ratio and Punnett square.
- Sex determination in humans: Father (XY) determines gender of child.
- **Practice**: Sketch the heart diagram and the Mendel $3:1$ cross chart.

Day 4 (Weekend): Chemistry Core — Salts, Metals & Carbon (16 Marks)

- **Target Chapters**: L.8 (Acids, Bases & Salts), L.27 (Metals & Non-metals), L.28 (Carbon Compounds)
- **Core Topics**:
- Preparation and 2 uses of: Bleaching Powder, Baking Soda, Washing Soda, Plaster of Paris.
- pH scale: Acids (< 7), Neutral (7), Bases (> 7).
- Reactivity series & displacement reactions.
- Carbon: Catenation, Tetravalency, Homologous series, and Cleansing action of soap (Micelles).
- **Practice**: Write down the 4 salt preparation chemical equations twice.

Day 5: Mechanics, Gravitation & Periodic Table (12 Marks)

- **Target Chapters**: L.10 (Force & Motion), L.11 (Gravitation), L.5 & L.6 (Atoms & Periodic Table)
- **Core Topics**:
- Newton's Second Law: $F = ma$ and momentum ($p = mv$).
- Universal Law of Gravitation: $F = G\frac{Mm}{r^2}$. Difference between g (acceleration due to gravity) and G (Universal constant).
- Mass (constant, scalar) vs Weight ($W = mg$, variable, vector).
- Modern Periodic Table: Periodic trends across periods (left to right) and down groups (atomic size, valency, metallic character).

Day 6: Ecology & Top 50 Objectives Blitz (42 Marks!)

- **Target Chapters**: L.29 (Natural Environment), L.30 (Human Impact), and all objective questions.
- **Core Topics**:
- 10% Law of Energy Transfer (Lindeman's rule) in food chain.
- Ozone layer depletion: Reaction of Chlorofluorocarbons (CFCs).
- Biomagnification of pesticides (DDT).
- Solve the entire file: `03_TOP_50_REPEATED_OBJECTIVES_MCQ_SOLVED.md`.

Day 7: Final Rehearsal & Exam Day Warm-Up

- **Activities**:
- Attempt the 85-mark `06_PREDICTED_MOCK_TEST_PAPER_2026.md`.
- Check your answers against the provided solutions.

- Read the `02_ALL_IN_ONE_SCIENCE_FORMULA_AND_CONCEPTS_SHEET.md` on exam morning.
- Exam Day Kit: Admit card, black/blue ballpoint pens, pencil, eraser, ruler (for ray diagrams & circuits).